

KAPIL BHATTARAI

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EDUCATION

M.S. in Agricultural & Biological Engineering

Jan 2025 – Present | GPA: 4.0/4.0

University of Florida, Gainesville, FL — UF Fellowship Recipient | Thesis Advisor: Dr. Gerrit Hoogenboom

B.S. in Agriculture

2017 – 2023

Agriculture and Forestry University (AFU), Rampur, Chitwan, Nepal

TECHNICAL SKILLS

Crop Modeling:	DSSAT v4.8, Model Calibration & Validation, Scenario & Sensitivity Analysis
Programming:	Python, R/RStudio, SQL, Fortran (learning), Git/GitHub, Shiny Apps
Data Science & ML:	Scikit-learn, TensorFlow/TFLite, Predictive Modeling, Time Series Analysis, Data Visualization
Geospatial:	QGIS, GeoPandas, Remote Sensing

EXPERIENCE

Graduate Research Assistant

Jan 2025 – Present

McNair Bostick Simulation Lab, Dept. of Agricultural & Biological Engineering, University of Florida — Advisor:
Dr. Gerrit Hoogenboom

- Conducting thesis research on developing a process-based model for lettuce growth under hydroponics settings.
- Developing enhanced graphical visualization tool (GB2) for DSSAT v4.8.6 using Python with modern UI and improved data processing capabilities.
- Building technical skills in Fortran and Python for crop model development and large-scale data analysis.
- Provided technical guidance and DSSAT training to visiting scholars in the Crop Simulation Laboratory.

Nursery Manager

Mar 2024 – Sep 2024

Plant Paradise — Kathmandu, Nepal

- Managed end-to-end nursery operations including plant propagation, production scheduling, inventory control, and retail sales.
- Drove digital marketing (social media, website updates) and applied data-driven sales planning to increase seasonal revenue.

Agriculture Intern

Aug 2022 – Aug 2023

Ragolski & Son's Ltd — Netiv HaAsara, Israel

- Operated precision growing systems in high-tech greenhouses, including coco-peat substrate management, drip irrigation scheduling, and controlled-environment monitoring.
- Gained hands-on experience in hybrid vegetable variety production and commercial cut-flower operations — managing planting schedules, spraying, harvesting, packing, and quality grading.

Undergraduate Researcher & Agriculture Intern

Feb 2022 – Jul 2022

Prime Minister Agriculture Modernization Project (PMAMP) — Chitwan, Nepal

- Precision nitrogen management in spring rice with SPAD, GreenSeeker & leaf color chart; SPAD-guided dosing cut applied N ~33% (120→80 kg/ha) with no yield loss (peer-reviewed, 2024).
- Delivered farmer training on mechanization, vermicomposting, and climate-smart agriculture practices.

SELECTED PROJECTS

DSSAT Graphical Visualization GB2

Python · DSSAT v4.8.6 · Data Visualization · Fortran | 2025 – Present

Enhanced graphical visualization tool for DSSAT crop model outputs, contributing to the global crop modeling research community (100+ countries).

Rogcheck — Crop Disease ID App

CNN · TensorFlow Lite · Java · Android

Android app for offline detection of 10 crop diseases across potato, pepper bell, and tomato using a convolutional neural network.

Personal Portfolio Website

Next.js · React · CSS Modules

Designed and built a full-stack portfolio site with dark/light theme, SEO metadata, and static export hosted at bhattaraikapil.com.np.

CERTIFICATIONS

Certificate in Biological Systems Modeling	University of Florida	<i>9 credit hrs</i>
DSSAT 2025 International Training	University of Georgia, Griffin, GA	<i>May 2025</i>
AI Fluency: Framework & Foundations	Anthropic	<i>2025</i>
Basic Image Classification with TensorFlow	Coursera Project Network	<i>Apr 2022</i>
Remote Sensing Image Acquisition, Analysis & Applications	University of South Wales	<i>Mar 2022</i>
Data Analysis with R Programming	Coursera	<i>Jun 2021</i>
Introduction to Machine Learning	Duke University	<i>Apr 2021</i>
Working with Geospatial Data in Python	DataCamp	<i>Apr 2021</i>
Getting Started with AWS Machine Learning	AWS	<i>May 2021</i>

PRESENTATIONS & EVENTS

DSSAT 2026 International Training — University of Georgia	Faculty	<i>May 2026</i>
Poster Presentation — University of Florida	Hydroponic Lettuce Growth Model	<i>Mar 2026</i>
Crop Physiology Training — Wageningen University & UF	1.5 ECTS Credit	<i>Nov 2025</i>
23rd DSSAT Development Sprint — University of Kentucky	Virtual Participant	<i>Aug 2025</i>
DSSAT 2025 International Training — University of Georgia	72 participants · 28 countries	<i>May 2025</i>
Poster Presentation — University of Florida	DSSAT CSM Performance Evaluation	<i>Mar 2025</i>
HiPerGator Supercomputer Facility Visit — University of Florida	UF Research Computing	<i>Mar 2025</i>
22nd DSSAT Development Sprint — University of Florida	In-person Participant	<i>Jan 2025</i>